Data Structures in Python

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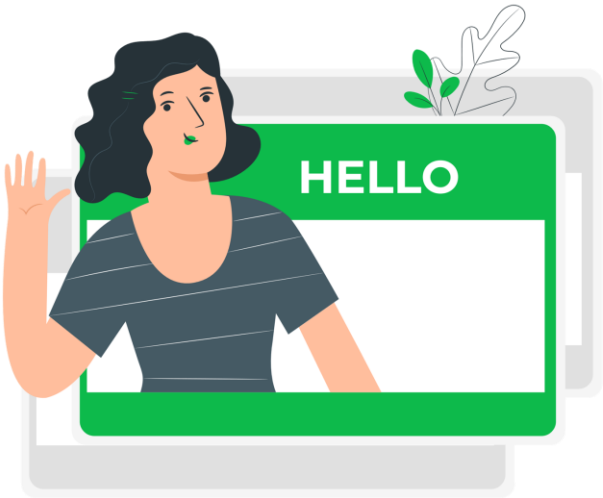
Agenda

1. Introduction to Loops
2. Types of Loop
3. Loop Control Statements (break, continue)
4. Scenario-Based Problem Solving using Loops
5. Nested Loops
6. Functions in Python
7. Local vs Global Variables
8. Introduction to Classes
9. Understanding the self Keyword



Unit Overview

Day 2 focuses on building **logical repetition, structured programming, and modular coding skills**. Learners will move from writing simple conditional programs to building reusable logic using loops and functions. Introduction to classes and the self keyword prepares learners for object-oriented programming.

Loops in Python

Why Loops?

- Avoid repetition of code
- Execute a block multiple times
- Automate repetitive tasks

Real Example:

- Printing salary slips for 100 employees
- Calculating total marks for multiple students

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Types of Loops

- for loop
- while loop

for Loop

```
for i in range(5):
print(i)
```

Explanation:

- range(5) → 0 to 4
- Loop runs 5 times

for Loop with List

```
names = ["Anu", "Raj", "Kiran"]

for name in names:

print("Hello", name)
```


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Types of Loops

while Loop

```
count = 1
while count <= 5:
    print(count)
    count += 1
```

Important:

- Condition-based loop
- Risk of infinite loop

Infinite Loop

```
while True:
print("Running...")
```


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Scenario-Based Problem Solving on Loops

Scenario 1 – Sum of Numbers)

Problem: Calculate sum of first N numbers

```
n = 5
sum = 0
for i in range(1, n+1):
sum += i
print("Total:", sum)
```


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Scenario-Based Problem Solving on Loops

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Scenario 2 – Factorial Program

)

```
num = 5
fact = 1
for i in range(1, num+1):
fact *= i
print("Factorial:", fact)
```


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Scenario-Based Problem Solving on Loops

Scenario 3 – Password Retry System

```
correct_password = "admin"
attempts = 3
while attempts > 0:
pwd = input("Enter password: ")
if pwd == correct_password:
print("Access Granted")
break
else:
attempts -= 1
print("Attempts left:", attempts)
```


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Nested Loops

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What is Nested Loop?

- Loop inside another loop
- Used for patterns, tables, matrix operations

Multiplication Table

```
for i in range(1, 6):
    for j in range(1, 6):
        print(i * j, end=" ")
    print()
```


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Functions in Python

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Why Functions?

- Reusability
- Better organization
- Avoid duplication

Function Syntax

```
def greet():

print("Welcome!")

greet()
```


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Functions in Python

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Function with Parameters

```
def add(a, b):
    return a + b
result = add(10, 5)
print(result)
```

Real Scenario – Salary Calculator

```
def calculate_salary(basic):
    hra = basic * 0.2
    da = basic * 0.1
    return basic + hra + da
print(calculate_salary(30000))
```


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Local vs Global Variables

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Local Variable

- Defined inside function
- Accessible only inside function

```
def test():
    x = 10
    print(x)
```

Global Variable

- Defined outside function
- Accessible everywhere

```
x = 20

def show():
    print(x)
```


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Local vs Global Variables

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Using global Keyword

```
count = 0

def increment():

    global count

    count += 1

increment()

print(count)
```


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Introduction to Classes & self Keyword

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What is a Class?

- Blueprint for creating objects
- Supports OOP concepts

Basic Class Example

```
class Student:
    def __init__(self, name):
        self.name = name
    def greet(self):
        print("Hello", self.name)
s1 = Student("Anita")
s1.greet()
```


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Practice Tasks

- Print even numbers from 1 to 50
- Generate multiplication table
- Create calculator function
- Create simple Student class

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Key Outcomes

By the end of Day 2, learners will:

- Write programs using for and while loops
- Solve real-world problems using iteration
- Create reusable functions
- Understand variable scope clearly
- Understand how self works inside classes

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Thank You!